SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	Shaik Azeem	**Reg. No.:**	192472307
Section / Batch:	CSE-AI	**Date:**	02/10/2026

Code of Conduct

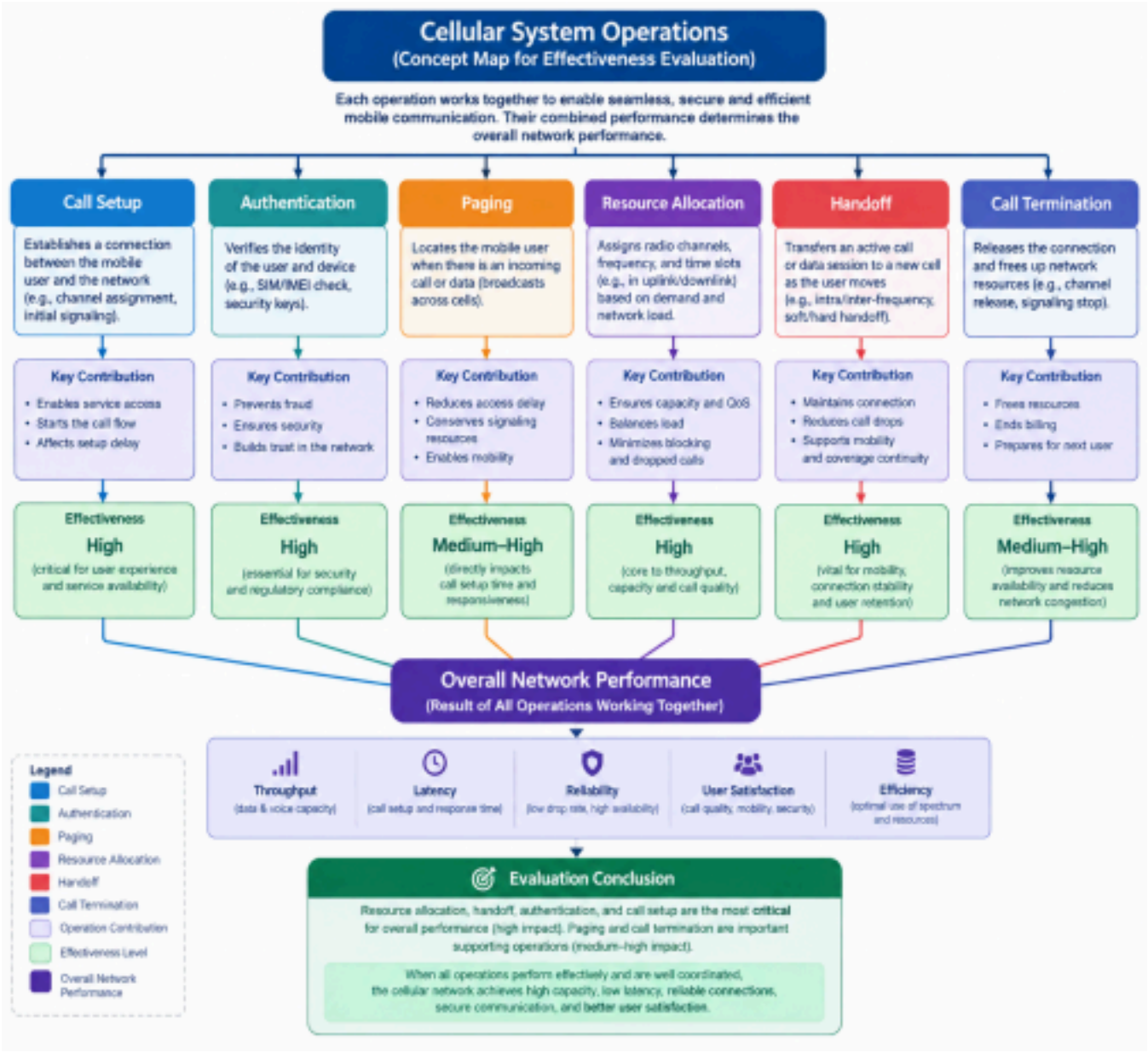
*I **Shaik Azeem(192472307)**(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the Student: **Shaik Azeem***

Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

1. **Construct a concept map that evaluates the effectiveness of various cellular system operations. Include call setup, authentication, paging, resource allocation, handoff, and call termination. Assess the contribution of each operation to overall network performance. Support your evaluation through logical concept relationships.**


Cellular System Operations
(Concept Map for Effectiveness Evaluation)
Each operation works together to enable seamless, secure and efficient mobile communication. Their combined performance determines the overall network performance.
Call Setup
Establishes a connection between the mobile user and the network (e.g., channel assignment, initial signaling).
Key Contribution
• Enables service access
• Starts the call flow
• Affects setup delay
Effectiveness
High
(critical for user experience and service availability)
Authentication
Verifies the identity of the user and device (e.g., SIM/IMEI check, security keys).
Key Contribution
• Prevents fraud
• Ensures security
• Builds trust in the network
Effectiveness
High
(essential for security and regulatory compliance)
Paging
Locates the mobile user when there is an incoming call or data (broadcasts across cells).
Key Contribution
• Reduces access delay
• Conserves signaling resources
• Enables mobility
Effectiveness
Medium–High
(directly impacts call setup time and responsiveness)
Resource Allocation
Assigns radio channels, frequency, and time slots (e.g., in uplink/downlink) based on demand and network load.
Key Contribution
• Ensures capacity and QoS
• Balances load
• Minimizes blocking and dropped calls
Effectiveness
High
(core to throughput, capacity and call quality)
Handoff
Transfers an active call or data session to a new cell as the user moves (e.g., intra/inter-frequency, soft/hard handoff).
Key Contribution
• Maintains connection
• Reduces call drops
• Supports mobility and coverage continuity
Effectiveness
High
(vital for mobility, connection stability and user retention)
Call Termination
Releases the connection and frees up network resources (e.g., channel release, signaling stop).
Key Contribution
• Frees resources
• Ends billing
• Prepares for next user
Effectiveness
Medium–High
(improves resource availability and reduces network congestion)
Overall Network Performance
(Result of All Operations Working Together)
Throughput
(data & voice capacity)
Latency
(call setup and response time)
Reliability
(low drop rate, high availability)
User Satisfaction
(call quality, mobility, security)
Efficiency
(optimal use of spectrum and resources)
Legend
Call Setup
Authentication
Paging
Resource Allocation
Handoff
Call Termination
Operation Contribution
Effectiveness Level
Overall Network Performance
Evaluation Conclusion
Resource allocation, handoff, authentication, and call setup are the most critical for overall performance (high impact). Paging and call termination are important supporting operations (medium–high impact).
When all operations perform effectively and are well coordinated, the cellular network achieves high capacity, low latency, reliable connections, secure communication, and better user satisfaction.

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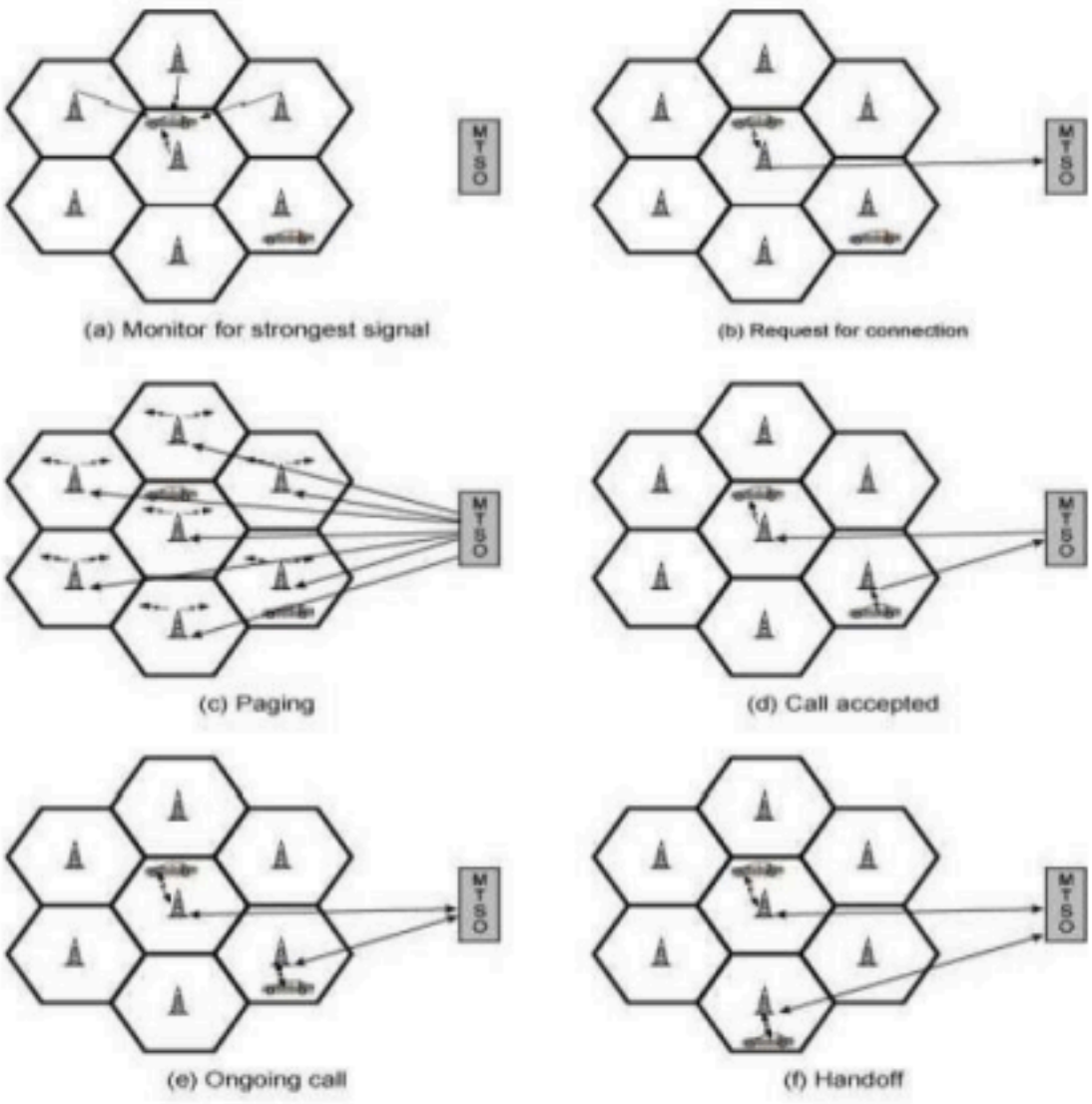


Fig 1.9 Call stages

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts

Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: _____	Signature: _____	Signature: _____

Date: _____ Date: _____ Date: _____